

CSE6324-SOFTWARE VERIFICATION & VALIDATION

Assignment

1. This is a GROUP assignment consisting of 3(min)- 4(max) students in a group.
2. Strictly NO individual work is allowed.
3. Grouping based on the same tutorial session. If your name is not listed in the Excel group list (which will be shared by the lecturer), you will get zero marks for the assignment.
4. Each group must select an existing open-source application from GitHub. You may choose from Web, Mobile, or IoT domains based on the following categories:
 - a. Web Applications: Systems such as e-commerce platforms, student course registration portals, library management systems or online chess games.
 - b. Mobile Applications: Functional apps like games, location-based services (GPS), or e-ticketing systems.
 - c. IoT Solutions: Projects involving heart rate monitoring, health indicators, or smart climate/temperature control
5. To ensure the project is suitable for Software Verification and Validation (V&V) activities, the selected GitHub repository must meet the following criteria:
 - a. **Documented Requirements:** The repository must contain a clear README.md or dedicated documentation file that outlines the functional requirements and specifications. Projects without sufficient documentation will be rejected.
 - b. **Testability:** The source code must be accessible and capable of being integrated with an automated testing or test management tool for the Assignment 2 demo.
 - c. **Local Build Verification:** Before requesting approval, groups must verify that the selected project can be successfully built and executed on a local machine.
6. The selected repository must be approved by the lecturer.
7. The assignment will be split into 2 different submissions. Assignment 1 will be assessed as a continuous assessment (25%), and Assignment 2 will be assessed as a Final Assessment (30%).

ASSIGNMENT 1**DATELINE: 4th MAY 2026 (WEEK 6)**

1. For Assignment 1, you're required to provide the following reports separately:
 - a. **Report 1: Software Requirements Specification** (minimum 10 functional and 5 non-functional requirements)
 - b. **Report 2: Test Plan** (include automated testing)
 - c. **Report 3: Test Design Specification**
 - d. **Report 4: Propose automated testing/Test management tool**
 - i. Suggest what automated tool/test management tool to use
 - ii. Discuss the advantages and potential pitfalls of using the tool, including ease of use, scalability, integration, and limitations.

ASSIGNMENT 2**DATELINE: 15 JUNE 2026 (WEEK 12)**

1. For Assignment 2, you're required to provide the following reports separately:
 - a. **Report 1: Test Automation Tool** (Demo – Provide a screenshot)
 - b. **Report 2: Test Cases**
 - c. **Report 3: Test Procedure Specification**
 - d. **Report 4: Test Log**
 - e. **Report 5: Test Summary Report**
 - f. **Report 6: Test Incident Report**
2. Video Presentation: Prepare a 3–5-minute video presentation explaining your findings, the script used in the automated tool, and a demo using the automated tool.

Report Submission Guidelines for Assignment 1 and Assignment 2.

1. Submit the assignment by the specified deadline.
2. Your report must be supported by suitable references such as journals, conference papers, or books (excluding textbooks and lecture slides). Website links may be included for tools (if applicable).
3. Use APA format for citations and references.
4. Each report must be prepared and uploaded as a separate document. Students are not allowed to combine all reports into one single file.
5. Documents to submit:
 - I. *Cover page (Include student IDs, names, topic, submission date, subject code & name, etc.)*
 - II. *Report content*
 - III. *References (if any)*

6. Submit the following table as a separate PDF file:

Student Name	ID	Task Completed	% of Involvement (TOTAL 100%)
...	...	...	

7. The Group Leader ONLY must submit all documents via Microsoft Form (details will be shared on Microsoft Teams/EBWise). One submission per group.
8. Turnitin report must be attached. Failure to do so will result in a 5-mark deduction.
9. A 5-mark deduction applies if the similarity index exceeds 10%.
10. A 10-mark deduction applies if the AI Turnitin index exceeds 20% (visible to the instructor only).
11. Late submissions will incur a 10-mark deduction per day from the overall mark.
12. No appeal will be considered unless supported by valid evidence in accordance with the examination process.

Assignment Rubric	
DESCRIPTION	MARKS
Assignment 1 (25%) - CONTINUOUS ASSESSMENT	
Software Requirements Specification	10
Test Plan	15
Test Design Specification	15
Test management tool	10
Total	50
Assignment 2 (30%) - FINAL ASSESSMENT	
Test Automation Tool (Functional Requirement) Demo	20
Test Cases	20
Test Procedure Specification	20
Test Log	10
Test Summary Report	10
Test Incident Report	10
Video Presentation	10
Total	100